Topological Data Analysis

Classification of Epileptic Seizure

SF2956 Topological Data Analysis
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1 Introduction

Topological data analysis (TDA) is a data analysis tool based on the assumption that data has shape and that the shape is meaningful. Fundamental methods in data analysis often rely on assumptions about the shape of the data, e.g. linearity using *Linear Regression* and convex clusters using *K-means clustering*. TDA can be seen as a general method of detecting different geometrical shapes of a data set; without any underlying assumptions, using the theory of homology. The use of TDA has shown promising results in detecting informative structures in large data sets (see Nicolau et al. (2011)). TDA provides a robust framework for analysing complex data sets without requiring high computational costs, as it can be used as a powerful tool for dimensionality reduction. However, the extent of using TDA as a tool in data analysis is still limited and therefore also the knowledge about its usefulness in different areas.

1.1 Problem formulation

This project investigates if patients having and not having an ongoing epileptic seizure can be distinguished by applying methods of persistant homology to a data set consisting of electroencephalograms (EEG) time series, recorded at the moment of a suspected seizure. An attempt to classify future observations is then made using features originating from the application of persistant homology in conjunction with k Nearest Neighbours (kNN).

1.2 Data set

The chosen data; *Epileptic Seizure Recognition*[1], contains time series of 4097 electroencephalograms (EEG) readings. The readings have been measured for over 23.5 seconds and on a total of 500 patients. One row in the data set corresponds to EEG readings of a patient during 1 second, thus every patient has 23 rows in the data set. As the 4097 readings of one patient is divided into 23 rows, each row contains 178 readings (corresponding to 178 columns). In addition to the 178 columns, there is one column holding the patient ID, and one column holding the label y which represents the condition class of the patient, $y \in \{1, 2, 3, 4, 5\}$. Out of these five condition classes, $y = 1$ corresponds to the patient having a seizure, whilst the other $y \in \{2, 3, 4, 5\}$ corresponds to patients not having a seizure. From this, we transformed the data into a binary response case, of which class 1 corresponds to patients having a seizure ($y = 1$), and class 0 corresponds to patients not having a seizure ($y \in \{2, 3, 4, 5\}$).

[1]The data can be obtained from
https://archive.ics.uci.edu/ml/datasets/Epileptic%2BSeizure%2BRecognition

2 Preliminaries

This section states the most important definitions from which the analysis is derived. It is not intended to be read in itself but rather referred to in the following sections, and can be used if any of the concepts are unfamiliar.

Definition 2.1 (Distances and pseudometrics)**.** Let X be a set. A distance on X is a function:

$$d : X \times X \to [0, \infty]$$

satisfying the following conditions for all x and y in X:

1. symmetry: $d(x, y) = d(y, x)$
2. reflexivity: $d(x, x) = 0$.

In addition, d is called a pseudometric if it also satisfies the triangle inequality:

3. $d(x, y) + d(y, z) \leq d(x, z)$.

Definition 2.2 (Partitions and discrete pseudometrics)**.** A partition of a set x_i is equivalent to the equivalence relation induced by a discrete pseudometric

$$d_t(x, y) := \begin{cases} 0, & \text{if } d(x, y) \leq t \\ \infty, & \text{otherwise} \end{cases}$$

where $d(x, y)$ is a pseudometric.

The following are some examples of pseudometrics on $\mathbb{R}^n$:
The **Euclidean metric**, also called the l_2 metric

$$d_2(x, y) = ||x - y|| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}, \tag{1}$$

the **Cityblock metric**, also called the l_1 metric

$$d_1(x, y) = ||x - y||_1 = |x_1 - y_1| + \cdots + |x_n - y_n|, \tag{2}$$

and the **Chebyshev metric**

$$d_\infty(x, y) = ||x - y||_\infty = max\{|x_1 - y_1|, \ldots, |x_n - y_n|\}. \tag{3}$$

The following is an example of a distance on $\mathbb{R}^n$:
The **Cosine distance**

$$d(x, y) = \begin{cases} 1 - \frac{x \cdot y}{||x|| \cdot ||y||}, & x \neq 0, y \neq 0 \\ \infty, & \text{otherwise.} \end{cases} \tag{4}$$

Definition 2.3 (Gromov-Hausdorff distance)**.** Let $\epsilon \in [0, \infty]$ and (X, d_X) and (Y, d_Y) be pseudometric spaces. The Gromov-Hausdorff distance between (X, d_X) and (Y, d_Y) is defined as

$$GH(X, Y) = \inf\{\epsilon \in [0, \infty] | (X, d_X) \text{ and } (Y, d_Y) \text{ are } \epsilon\text{-close}\}. \quad (5)$$

Definition 2.4 (Extension)**.** Let $\mathcal{D}(X)$ denote the set of all distances on X, and let $\mathcal{D}(\Delta[X])$ denote the set of all distances on all non-empty finite subsets of X. Then, an extension ψ is a function $\psi : \mathcal{D}(X) \to \mathcal{D}(\Delta[X])$.

Thus, ψ is a function from distances between points in X to distances between subsets of X. The following is an example of an extension:
Let d be a distance on X. For non-empty finite subsets A and B of X, we define the **single linkage** extension of d as

$$sd(A, B) = min\{d(a, b) | a \in A, b \in B\}. \quad (6)$$

Definition 2.5 (Simplicial complex)**.** Let $\mathcal{U}$ be our universe of interest. A simplicial complex is a collection K of some finite non-empty subsets of $\mathcal{U}$ that satisfies that if $\sigma \subset \mathcal{U}$ is in K, then any non-empty subset of σ is also in K.

Definition 2.6 (Vietoris-Rips complexes)**.** Let $\mathcal{U}$ be our universe of interest and d be a distance on $X \subset \mathcal{U}$. For $t \in [0, \infty)$ a Vietoris-Rips complex of (X, d) at scale t, $VR_t(d)$, is defined to be the collection of the finite non-empty subsets $\{x_0, ..., x_n\} \subset X$ such that $d(x_i, x_j) \leq t$ for any i and j.

Definition 2.7 (Homology function H_n)**.** The homology function H_n, or the n-th homology of X, is defined as

$$H_n(X, K) = \ker(\delta_n)/\text{im}(\delta_{n+1}). \quad (7)$$

For a more extensive definition of the homology function, see Chachólski (n.d.).

Remark (Homology function H_n). The homology function H_n is used when converting finite simplicial complexes into finite dimensional vector spaces. Homology measures how many more cycles than boundary elements there are in a set. One can interpret H_0 as the number of components in the data set, H_1 as the number of one-dimensional holes in the data set and H_2 as the number of two-dimensional holes in the data set.

Definition 2.8 (Piece-wise constant function, PCF)**.** A function $f : [0, \infty] \to (-\infty, \infty)$ is called piece-wise constant if there is a finite number of elements $0 = a_0 < a_1 < ... < a_n \in [0, \infty]$ for which the restrictions of f to the right open intervals $[a_0, a_1), ..., [a_{n-1}, a_n), [a_n, \infty)$ are constant functions.

Definition 2.9 (Interleaving distance)**.** The interleaving distance $d_{\bowtie}$ is a metric on the set of PCFs. Let f and g be piece-wise constant function, then

$$d_{\bowtie}(f,g) = \begin{cases} \infty, & \text{if } \{v | f(x) \geq g(x+v) \text{ and } g(x) \geq f(x+v) \text{ for any } x\} = \emptyset \\ \inf\{v | f(x) \geq g(x+v) \text{ and } g(x) \geq f(x+v) \text{ for any } x\}, & \text{else.} \end{cases} \tag{8}$$

Definition 2.10 (k Nearest Neighbours, k NN)**.** k Nearest Neighbours, kNN, is a non-parametric method which can be used for classification. In kNN, an object is classified by a plurality vote of its k nearest neighbors, as the object is assigned to the class most common among them. The k nearest neighbors is based on some predefined similarity measure, i.e. some type of distance function.

3 Analysis pipeline

In the context of TDA, each data point x_i in a data set $X = \{x_i\}_{i=1}^{N}$ is given in the form of a point cloud. Each data point is thus on the form $x_i \in \mathbb{R}^{m \times n}$ corresponding to n feature vectors with dimension m.

TDA enables one to quantify shape in data. In order to allow this; a notion of dissimilarity between feature vectors has to be introduced in the form of a distance, resulting in a distance space. A distance space $\mathcal{D}(x_i) = (x_i, d)$ is defined on a data point x_i by introducing a distance function $d : \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}$ (2.1) on the feature vectors. The choice of distance function d has an important impact on the analytical process in the sense that if d is chosen as a pseudometric, the upcoming methods are guaranteed to be stable. This will be further enlightened later.

Two data points x_i and x_j can be compared in a topological manner by comparing the corresponding distance spaces $\mathcal{D}(x_i)$ and $\mathcal{D}(x_j)$. One method (theoretically the best one) of comparing two distance spaces is the *Gromov-Hausdorff* distance $\mathrm{GH}(\mathcal{D}(x_i), \mathcal{D}(x_j))$, defined in (2.3). This loosely corresponds to how well the two distance spaces fit into each other. A virtue of GH is that it very accurately quantifies how dissimilar two distance spaces are. This is useful in the context of classification where for example distance spaces on data points from the same class are expected to be similar. However, the preciseness of GH comes with a drawback. The number of computations needed for the distance scales with the factorial, making it computationally infeasible. Simplification of the distance space is thus crucial and one of the fundamentals in TDA.

In this project, the objective is to simplify distance spaces $\mathcal{D}(x_i)$ by finding a topological representation of them. This could be achieved through two methods, standard hierarchical clustering and persistant homology.

3.1 Standard hierarchical clustering

A possible way of producing a simplified representation of the distance space (x_i, d) is to consider the partitions of x_i that a discrete pseudometric induces at different scales t (2.2). By further restricting the representation to only consider the number of blocks in a partition at a scale t, we end up with a process called *stable rank*,

$$\widehat{\mathbf{rank}} : (x_i, d) \to \mathbf{pcf}_d. \tag{9}$$

The first step involved in producing such representation is the choice of an *extension* ψ, (2.4). Using ψ, it is possible to construct a corresponding dendrogram, which can be thought of as a time series of sets. The sets at different times t, are simply the partition induced by the discrete pseudometric at scale t. This representation can now be further simplified by converting it to a time series of vector spaces $V(t)$, where the dimension at every t corresponds to the number of sets in the dendogram at time t.

It is now possible to use the *dimension* of $V(t)$ and represent it as a *piecewise constant function*, PCF (2.8). This representation now allows us to compare different distance spaces and perform statistics.

3.2 Persistant homology

A distance space comprises of more complex geometrical structures than partitions, such as cycles and holes. Persistant homology; in contrast to standard hierarchical clustering, also enables one to quantify the more complex geometrical structures through PCFs. The process of finding

$$\widehat{\mathbf{rank}} : (x_i, d) \to \mathrm{pcf}_d, \tag{10}$$

thus takes a different form. The distance space (x_i, d) is transformed to a *simplicial complex* parameterised by $[0, \infty)$ by applying *Vietoris-Rips*, $\mathrm{VR}_\bullet(d)$, (2.6). From this geometrical representation of (x_i, d) one can identify the number of partitions, cycles etc. at different scales t, by applying the homology function H_n (2.7). This can then similar to before be represented by a time series of vector spaces $V(t)$. Typically, we would choose to use the homology functions H_0, H_1 and in some cases H_2, corresponding to partitions, one- and two dimensional cycles respectively.

Further, it is known from the theory that the time series of vector spaces $V(t)$ is isomorphic to a direct sum of bars, i.e. $V(t) \simeq \oplus_{i=1}^{n} [a_i, b_i)$. This allows us to construct the $\mathbf{pcf}_d$ representation by defining

$$\widehat{\text{rank}}V(t) = |\{i \mid b_i - a_i \geq t\}| \,. \tag{11}$$

It is important to note that the process of going from a distance space to a stable rank representation, is only guaranteed to be continuous when the distance is chosen as a pseudometric. In addition, the use of a pseudometric also guarantees that if an error ϵ is added to our data x, then there is a 2ϵ-equivalence between the parametrized vector spaces created by applying persistant homology to x and $x + \epsilon$. For this reason it is important to be cautious when using distances which are not pseudometrics.

4 Methodology

In this project, the analysis will be restrained to the homologies H_0 and H_1, using persistent homology (3.2). In this process, there are a number of entities which need to be specified. The extent of this analysis will cover a preprocessing step of the input data in which we will look at the effects of using different feature vectors, referred to as point cloud transformation. Furthermore, four different distances will be considered. In order to decide which point cloud transformation and distance to use a greedy approach will be taken. For example, the performance of the four different metrics will be compared to each other whilst holding the rest of the model constant. The distance with the best performance will be chosen for the final model, which is to be used in the classification of new data. This approach is taken to reduce the number of model permutations that needs to be evaluated; however, it is important to note that it only enables a local optimum.

4.1 Preprocessing of data

The chosen data set $\mathcal{T}$ is given on the form

$$\mathcal{T} = \{(y_i, x_i)\}_{i=1}^{11500}, \qquad x_i \in \mathbb{R}^{178},\, y_i \in \{1, 0\},\, \forall i, \tag{12}$$

where x_i corresponds to features of data point i, and y_i define the class of which x_i belongs to. We have earlier stated that class $y = 1$ corresponds to patients having had a seizure and vice versa.

In order to enable the application of persistent homology of dimensions larger than 0, the feature vector $x_i \in \mathbb{R}^{178}$ have to be transformed into a point cloud. In this analysis, $x_i \in \mathbb{R}^{2\times 89}$, will be considered. To achieve this

form, 3 different types of points cloud transformations is investigated; Low correlation-, High correlation- and Randomized points cloud. For each of the different point clouds, we will compute the average interleaving distance between class 1 (patients having seizure) and class 0 (patients not having seizures), for the average stable ranks H_0 and H_1. From this, we conclude which of the point cloud to use, hypothesising that the interleaving distance is related to how well the stable rank representation can separate the two classes.

The results is presented in section (5) and are based on computations where the Euclidean distance is used for computing the stable ranks. However, we carried out the same computations based on the Cityblock distance and the Chebyshev distance, which led to the same conclusion.

4.1.1 Low correlation point cloud

This transformation pairs features together such that the pairwise correlation is minimized. The hypothesis is that this potentially could lead to a large separation of the feature vectors in $\mathbb{R}^2$, allowing for a larger difference in geometrical shapes between the two classes.

4.1.2 High correlation point cloud

This transformation pairs features together such that the pairwise correlation is maximized. This is done as a complement to the low correlation transformation to further understand the effects of pairing features by correlation. The hypothesis of this point cloud is that is will perform worse in separating the two classes since the feature vectors in $\mathbb{R}^2$ will inhibit a linear shape for both classes.

4.1.3 Randomized point cloud

This transformation randomly assigns data points into pairs. This is mostly considered as a sanity check for whether the correlation-pairing has any effect on the separation of the two classes.

4.2 Choice of distance space

The performance of the Euclidean metric, Cityblock metric, Chebychev metric and Cosine distance (1-4) will be investigated. The three former are pseudometrics whilst the Cosine distance only fulfills the criteria for being a distance (2.1).

4.3 Classification using kNN

The classification of new data is performed using kNN (2.10) on the stable rank feature representations of the data, with the interleaving distance chosen as similarity measure. The classification use a training sample of 200 EEG-time series of each class ($N = 400$), and a test sample of 50 unseen EEG-time series of each class. As one wants the number of total votes to be odd, k was chosen as $k = \sqrt{N} \approx 21$. The average prediction accuracy was calculated on five rounds, where each run was done with new train and test data.

5 Results

Throughout this section, figures will be presented. In these figures, the blue graph corresponds to class 1 (patients having a seizure) and the red graph corresponds to class 0 (patients not having a seizure).

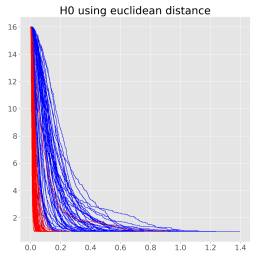
5.1 Choice of point cloud representation

The results of calculating the interleaving distance between the average stable ranks H_0 and H_1 for each class is presented in Table (1). Here, it can be seen that the the interleaving distance was the highest between the two classes when using the low correlation point cloud.

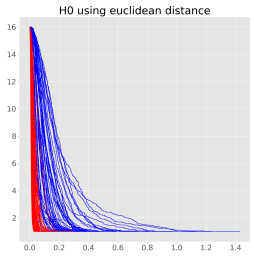
Feature clustering:	$d_{\bowtie}$ for H_0	$d_{\bowtie}$ for H_1
High correlation	0.95	0.21
Low correlation	1.07	0.25
Randomized	0.94	0.16

Table 1: Interleaving distance between the average stable ranks H_0 and H_1, using 50 data points of each class.

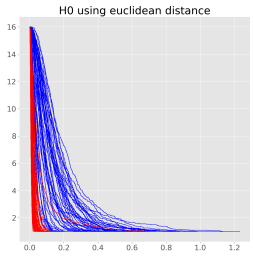
Furthermore, Figure (1) presents the stable rank PCFs and Figure (2) presents the average stable rank PCFs.



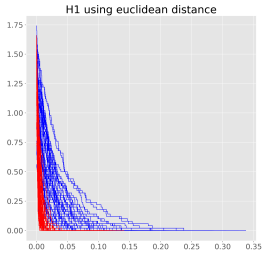
(a) High correlation



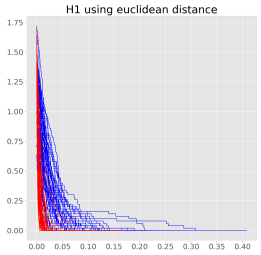
(b) Low correlation



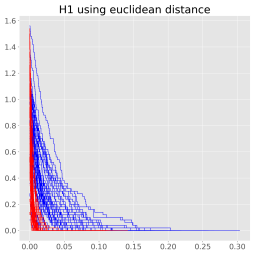
(c) Randomized



(d) High correlation

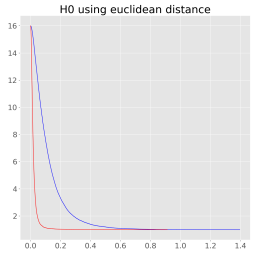


(e) Low correlation

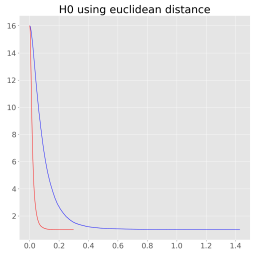


(f) Randomized

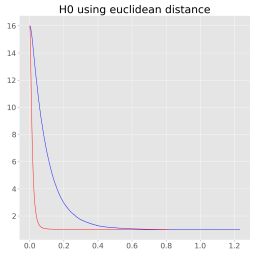
Figure 1: Stable rank representation H_0 and H_1 for different point clouds.



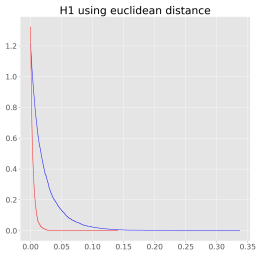
(a) High correlation



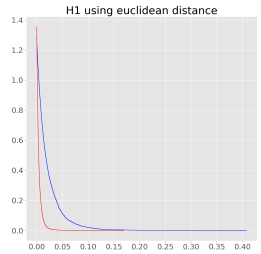
(b) Low correlation



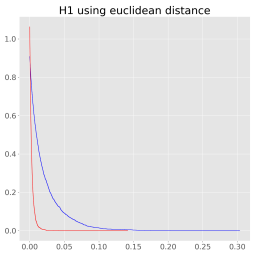
(c) Randomized



(d) High correlation



(e) Low correlation



(f) Randomized

Figure 2: Average stable rank representation H_0 and H_1 for point clouds.

As we wanted to choose the point cloud transformation that maximise the interleaving distance between the two classes, we chose the low correlation point cloud.

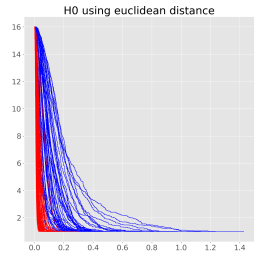
5.2 Choice of metric

The results of the interleaving distance between the average stable ranks for H_0 and H_1, based on different distances, is given in Table (2).

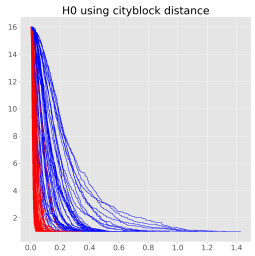
Distance:	$d_{\bowtie}$ for H_0	$d_{\bowtie}$ for H_1
Euclidean	1.13	0.24
Cityblock	1.08	0.35
Chebychev	0.84	0.27
Cosine	0.22	0.39

Table 2: Interleaving distance between the average stable ranks for H_0 and H_1 using 50 data points of each class, evaluated for different metrics.

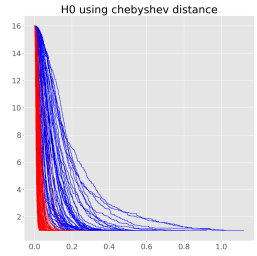
Furthermore, the stable ranks of H_0 is shown in Figure (3), and H_1 in Figure (4). The average stable rank of H_0 is shown in Figure (5) and the average stable rank of H_1 is shown in Figure (6).



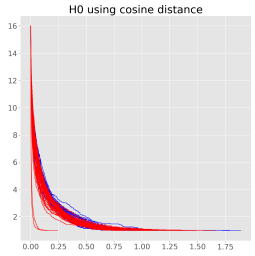
(a) Euclidean distance



(b) Cityblock distance

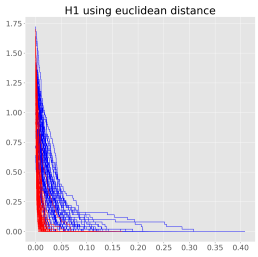


(c) Chebychev distance

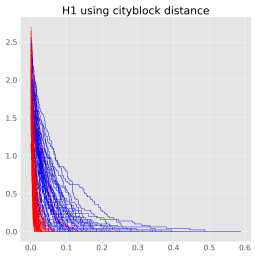


(d) Cosine distance

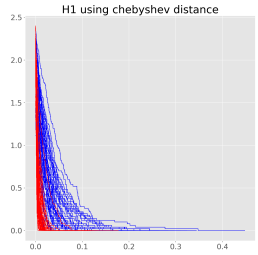
Figure 3: Stable rank representations of H_0 using different distances.



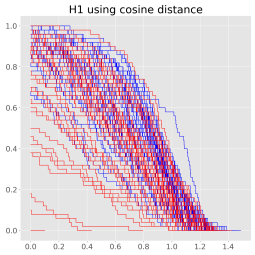
(a) Euclidean distance



(b) Cityblock distance

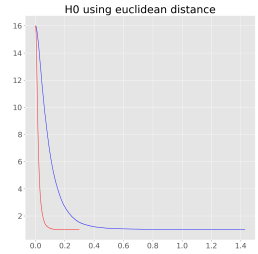


(c) Chebychev distance

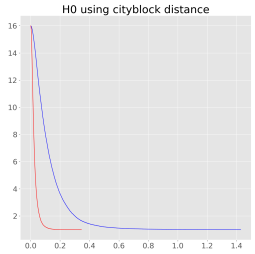


(d) Cosine distance

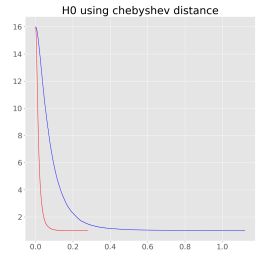
Figure 4: Stable rank representations of H_1 using different distances.



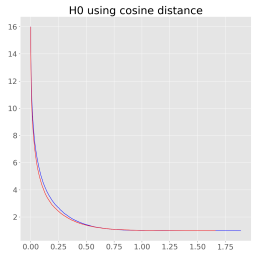
(a) Euclidean distance



(b) Cityblock distance

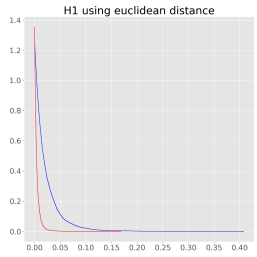


(c) Chebychev distance

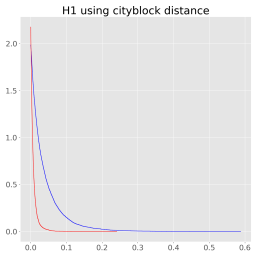


(d) Cosine distance

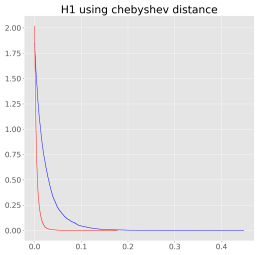
Figure 5: Average stable rank representation of H_0 using different distances.



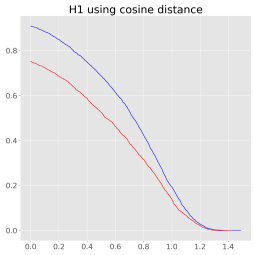
(a) Euclidean distance



(b) Cityblock distance



(c) Chebychev distance



(d) Cosine distance

Figure 6: Average stable rank representation of H_1 using different distances.

From Figure (4), it is clear that the cosine distance performs poorly in separating the classes in H_1, although the interleaving distance between the averages is relatively large (compare with Figure (6)). In Figures (3-6) it can be seen that the other three metrics seem to distinguish well between the classes, both in H_0 and H_1.

Looking at the interleaving distances presented in Table (2), it seems as the Euclidean- and the Cityblock distance yields the largest separation of the classes, for both homologies. The choice of the best metric from these results is not obvious, but we argue that the Cityblock metric is the best alternative, due to its large average separation in H_1.

5.3 Classification using kNN

From the results obtained above, we used the low correlation point cloud transformation of the samples, along with the Cityblock metric for computing the homologies. The average prediction accuracy of the kNN classification is presented in Table (3). Here, it can be seen that the classification yielded a high accuracy on the test set.

Homology:	H_0	H_1	H_0 and H_1
Prediction accuracy	0.944	0.936	0.948

Table 3: Average prediction accuracy (out of 5 runs) of classifying 50 new EEG-time series from each class.

6 Conclusion/Discussion

From the results, we conclude that the stable rank representation of our dataset $\mathcal{T}$ shows promising results in distinguishing epileptic from non-epileptic EEG recordings. In particular, when using the Cityblock distance for computing the homologies, the interleaving distances for the stable ranks H_0 and H_1 yields the most distinct results. In addition, we conclude that we are able to make accurate predictions of new data, by using tools from TDA along with the kNN classification method.

It should be pointed out that there is a large difference in magnitude between the classes. The EEG values are in general considerably larger during an epileptic seizure. Thus, one would assume that most classification methods would perform well in separating the classes.

As for future investigations, one could focus more on standard hierarchical clustering. Here, one could compute dendrograms obtained for different extensions such as the complete-, average- and ward linkage function. Then, one could compare each corresponding bar decomposition with the results we obtained for the stable rank representation for H_0. From this, one could perhaps draw further conclusions.

One could also verify the robustness of the method by introducing more noise in the data. As this is a data set containing real-world values, we expect it to already include some noise. However, the data is already quiet well separated which makes it a relatively simple classification task. If one would to increase the noise in the data, we would still expect the results to be stable. As earlier stated, since the Cityblock distance is a pseudometric, the TDA pipeline would guarantee the method to be robust. Though, one could perform an experiment to verify this.

References

Chachólski, W. (n.d.), 'Notes'.

Nicolau, M., Levine, A. J. & Carlsson, G. (2011), 'Topology based data

analysis identifies a subgroup of breast cancers with a unique mutational profile and excellent survival', *Proceedings of the National Academy of Sciences* **108**(17), 7265–7270.